

Do Stock Prices Move Too Much to Be Justified by Subsequent Changes in Dividends?

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ABSTRACT

Shiller asks whether the volatility of real stock prices is consistent with the standard efficient-markets present-value model, under which price equals the rationally expected present value of future real dividends. Rather than testing return predictability directly, he derives a variance-bounds inequality: because the ex-post rational price — the present value computed from dividends actually realized — equals the true price plus an orthogonal forecast error, its variance must be at least as large as the variance of the true price. Using detrended real Standard & Poor's data from 1871–1979 and a modified Dow Jones series from 1928–1979, he finds the inequality violated by an order of magnitude: actual prices are between six and thirteen times more volatile than the ex-post rational benchmark permits. Reconciling this with efficiency would require real discount rates fluctuating far beyond any plausible range. The paper does not conclusively prove irrationality — a claim later contested on statistical grounds — but it reoriented asset-pricing research toward the joint behavior of prices, dividends and discount rates, and opened the empirical study of excess volatility.

1. Introduction

PAPER Shiller's 1981 paper is not, on its surface, a test of return predictability, which was the dominant efficient-markets testing strategy of the 1970s following Fama (1970). It is instead a test of a stronger and more structural claim: that the observed level of a stock price is what a rational, fully-informed forecaster using the correct model would have set it to, given the information available at the time. The paper's device for testing this claim is to compare the volatility of the actual price series to the volatility of a constructed benchmark — the *ex-post rational price*, computed by discounting the dividends that were actually subsequently paid. If markets price shares as the present value of rationally expected future dividends, the actual price cannot be more volatile than this benchmark; it can, at most, be a smoothed version of it, since a rational expectation is by construction less variable than the realization it forecasts.

PAPER The finding is that this bound is violated by a wide margin in both of the two long time series examined. Real, detrended stock prices between 1871 and 1979 fluctuate roughly six times more than the ex-post rational benchmark for the Standard & Poor's series, and more than thirteen times more for the modified Dow Jones series over 1928–1979. The gap is not a modest statistical anomaly that a change of estimation window would erase; it is an order-of-magnitude discrepancy, and the paper spends much of its length trying, and failing, to reconcile it with reasonable auxiliary assumptions about time-varying discount rates or fat-tailed information arrivals.

CRITIQUE The question matters because the present-value relation the paper tests is not one asset-pricing model among many: it is the pricing identity that underlies essentially every discounted-cash-flow valuation exercise in finance, and a strict implication of no-arbitrage under rational expectations once one restricts attention to models with a well-defined discount rate. A demonstrated violation, if it survives methodological scrutiny, is a strike against the joint hypothesis of rational expectations and a reasonably stable discount rate — not against some auxiliary trading-rule profitability criterion, which is the weaker notion of efficiency that most earlier tests examined. This is why the paper, together with LeRoy and Porter's contemporaneous and methodologically related 1981 *Econometrica* paper, is usually credited with opening the modern literature on excess volatility, time-varying discount rates, and, eventually, behavioral asset pricing.

INFERENCE What the paper established, and what it merely suggested, turned out to require nearly a decade of subsequent econometric work to disentangle, because the variance-bounds test itself carries statistical assumptions — principally stationarity of the underlying price and dividend series — that are difficult to satisfy with detrended but still trending macro-financial time series. This review follows that arc: the mechanism first, on its own terms; then the specific statistical objections that followed; and finally the place the paper now occupies in a literature that has moved from “are prices too volatile” to “what combination of time-varying risk premia and investor psychology can account for the volatility that is there.”

2. Research Problem and Hypothesis

2.1 Research Question

PAPER The precise question is not “are stock markets efficient,” which is too coarse to be tested. It is: *conditional on the standard constant-discount-rate present-value model of stock prices, is the variance of the observed real price series consistent with the variance of the present value of the dividends that were subsequently, actually paid?* This is a second-moment restriction on the joint distribution of an observable price series and an observable dividend series, derived from the assumption that price equals the mathematical conditional expectation, given time- t information, of a well-defined random variable (the ex-post rational price). It does not require observing investors' information sets directly, forecasting models, or trading strategies — only the realized dividend stream, which sidesteps the joint-hypothesis problem of needing to specify the market's exact information set.

2.2 Motivation

PAPER By 1980, tests of weak-form and semi-strong-form efficiency (serial correlation in returns, event studies, filter rules) had repeatedly failed to reject efficiency, and the profession's working consensus, summarized in Fama (1970), was that markets priced securities correctly relative to available information. But those tests share a structural weakness: they detect whether *changes* in prices are predictable, which is a much weaker requirement than whether the *level* of price is consistent with fundamentals. A price series can follow an unpredictable random walk and still be persistently and dramatically detached from any reasonable measure of fundamental value, provided the detachment itself evolves unpredictably. Shiller's contribution is to design a test that speaks to the level question rather than the innovation question.

2.3 Existing Literature

PAPER The present-value framework itself is not new to this paper — it descends from the Gordon (1962) growth model and from Williams' (1938) dividend-discount theory of intrinsic value, and the martingale/random-walk characterization of efficient prices follows Samuelson (1965) and Fama (1970). What was missing, prior to 1981, was a way to confront the present-value model with data on the *level* of volatility rather than the *serial correlation* of returns. LeRoy and Porter, working independently and publishing in the same year, arrived at a closely related test using corporate earnings rather than dividends, which situates the two papers as a coordinated methodological turn in the literature rather than an isolated result.

2.4 Research Gap

INFERENCE The gap, stated in the paper's own terms, is that no existing test could distinguish a market that prices shares at the rational present value of dividends from a market whose prices deviate from that present value in ways that happen to be statistically unpredictable period to

period — because unpredictability of returns is consistent with both. A variance-bounds test closes part of that gap by imposing a restriction that a purely martingale-based test cannot: it uses the entire realized dividend path, not just the one-period-ahead conditional mean.

2.5 Contribution

CRITIQUE The claimed contribution is a new, more demanding efficiency test and an empirical finding that the standard model fails it by a wide margin on more than a century of U.S. equity data. This contribution is genuinely demonstrated for the specific null hypothesis tested — a constant real discount rate, geometric-declining-weight present-value model with rational expectations. It is not, and the paper is reasonably careful not to claim, a demonstration that markets are irrational in any welfare-relevant or exploitable sense; the paper's own robustness section already anticipates that time-varying discount rates could in principle rescue the model, and merely argues that the required variation is implausibly large by the standards known in 1981. Whether that plausibility judgment survives later work on time-varying risk premia is examined in Sections 7 and 9 below.

3. Theoretical Framework and Mechanism

PAPER The derivation rests on three primitives: a constant one-period real discount rate, a terminal (transversality) condition ruling out rational bubbles, and rational expectations. Let P_t denote the real, ex-dividend stock price at time t , D_t the real dividend paid at t , and r the constant required real return, with $\gamma = 1/(1+r)$.

3.1 The efficient-markets present-value model

$$P_t = \sum_{k=0}^{\infty} \gamma^{k+1} E_t[D_{t+k}] \quad (2)$$

Price is the sum of all future dividends, each discounted back to the present at the constant rate r , and each replaced by its mathematical expectation conditional on the information available at time t , $E_t[\cdot]$. This is the standard constant-discount-rate dividend-discount identity; it is the pricing equation implied by no-arbitrage whenever agents are risk-neutral, or, more generally, whenever the equilibrium stochastic discount factor happens to be non-stochastic.

3.2 The ex-post rational price

$$P_t^* \equiv \sum_{k=0}^{\infty} \gamma^{k+1} D_{t+k} \quad (4)$$

The ex-post rational price replaces each expected future dividend in equation (2) with the dividend that was actually, subsequently paid. It is not observable in real time — it requires the entire future path of dividends — but it is computable after the fact from realized data, which is exactly what

makes the test operational using a long historical sample. Because P_t^* is built from realizations rather than forecasts, it inherits all the variability actually present in the dividend stream, undamped by any averaging or expectation operator.

3.3 The forecast-error decomposition

Define the ex-post forecast error $u_t \equiv P_t^* - P_t$. Substituting (2) and (4),

$$u_t = \sum_{k=0}^{\infty} \gamma^{k+1} (D_{t+k} - E_t[D_{t+k}]) \quad (3)$$

u_t is a sum of pure dividend-forecast innovations, each of which is by construction orthogonal to time- t information — including P_t itself, since P_t is a function only of time- t expectations. Rational expectations is doing all the work in this step: it guarantees $E_t[u_t] = 0$ and $\text{Cov}(P_t, u_t) = 0$.

3.4 The variance-bounds inequality

Because $P_t^* = P_t + u_t$ with $\text{Cov}(P_t, u_t) = 0$, variances add:

$$\text{Var}(P_t^*) = \text{Var}(P_t) + \text{Var}(u_t) \Rightarrow \text{Var}(P_t) \leq \text{Var}(P_t^*) \quad (1)$$

EVIDENCE Equation (1) is the entire testable content of the paper stated as one line: the standard deviation of the true, rationally-priced series can never exceed the standard deviation of the perfect-foresight benchmark constructed from what dividends actually turned out to be, since a conditional expectation is a variance-reducing (orthogonal) projection of the object it forecasts. This is a completely general statistical fact — it holds regardless of the distribution of dividends, and in particular does not require normality, so an appeal to fat tails or occasional large information shocks cannot, on its own, rescue a violation of (1); a more variable u_t only ever pushes $\text{Var}(P_t^*)$ up, never down, relative to $\text{Var}(P_t)$.

P_t	real, detrended stock price at time t — the object rational pricing is claimed to explain
P_t^*	ex-post rational price — present value of the dividend path actually realized after t
D_t	real dividend paid at time t , detrended by the same long-run growth factor as P_t
r, γ	constant required real return and its discount factor $\gamma=1/(1+r)$, estimated from the sample average dividend-price ratio
u_t	ex-post forecast error, orthogonal to time- t information under rational expectations

3.5 What the derivation assumes, and what it predicts

CRITIQUE Three assumptions carry the full weight of equation (1), and each is a candidate point of failure examined later in this review: (i) the discount rate r is constant over the sample, so that the orthogonality of u_t is not contaminated by a time-varying weighting scheme correlated with dividend news; (ii) the terminal value of the discounted price sequence vanishes, ruling out rational bubbles that would add a further, non-fundamental component to P_t without violating rationality; and (iii) P_t and D_t , once detrended, are covariance-stationary, since sample variances are only meaningful, unbiased estimates of population variances under stationarity. The

prediction that follows, tested directly in Section 5, is the inequality $\sigma(P_t) \leq \sigma(P_t^*)$; the test compares sample standard deviations of the two detrended series computed from the same historical sample.

4. Data and Empirical Strategy

4.1 Identification: what makes the test operational

PAPER The identifying move is to substitute realized dividends for expected dividends in constructing P_t^* , which requires knowing the entire future dividend path from t to the end of the sample. Since the sum in equation (4) is infinite but geometrically discounted, Shiller truncates it at the end of the available sample and closes the series with a terminal-value proxy equal to the actual final detrended price, discounted back — a standard and low-influence approximation given how quickly γ^k decays at plausible real discount rates. The discount rate r itself is not estimated by a separate econometric procedure; it is set equal to the sample average detrended dividend–price ratio (0.0452 for the S&P sample, 0.0429 for the Dow sample), consistent with a steady-state version of equation (2) applied to the sample means.

4.2 Detrending

PAPER Real aggregate dividends grow over the long run with the economy, which means neither P_t nor D_t is stationary in levels; both are divided by a common exponential growth-trend factor before any variance comparison is made, so that the series analyzed are detrended real price and detrended real dividends. This step is necessary for the variance-bounds logic to be meaningful at all — comparing raw variances of two trending series would be comparing quantities dominated by the trend rather than by the deviations from it that the model is actually about — but it is also, as later critics point out, exactly the step that reintroduces the residual nonstationarity concerns addressed in Section 7.

4.3 Sample and sources

Table 1. Sample construction

Series	Sample period	Price index	Dividend source	Deflator
Data Set 1	1871–1979	Standard & Poor's Composite Stock Price Index	S&P Composite dividends per share, annual	Producer/wholesale price index
Data Set 2	1928–1979	Modified Dow Jones Industrial Average	Dow constituent dividends, annual, spliced for composition changes	Producer/wholesale price index

Source: Shiller (1981), Table I and accompanying data appendix; series description reconstructed from the paper and standard secondary treatments (Campbell, Lo and MacKinlay, 1997, ch. 7).

PAPER The two samples are deliberately different in length and construction — the S&P series reaches back to 1871, giving over a century of annual observations, while the Dow series is shorter but independently constructed and index-composition-adjusted — precisely so that the central finding does not depend on one index methodology or one historical period. Both real series are annual, end-of-year, single-country (U.S.) aggregate equity indices; the unit of observation is the calendar year.

4.4 Data limitations and candidate biases

- **Trend-estimation bias.** The exponential growth trend used for detrending is itself estimated from the same finite historical sample it is then used to detrend, which can mechanically induce excess variance in the detrended series even under the null — the central statistical objection raised by Flavin (1983) and Kleidon (1986), discussed in Section 7.
- **Terminal-value truncation.** Approximating the post-sample dividend tail with the actual terminal price rather than a fully independent forecast introduces a small end-of-sample sensitivity, though the geometric discounting makes this a second-order concern for a century-long sample.
- **Index-composition changes.** The Dow series requires splicing across changes in index membership, which is a known source of measurement error in long historical indices, partially mitigated by cross-checking against the independently constructed S&P series.
- **Constant discount rate.** Using the full-sample average dividend yield as a single, time-invariant r is itself a maintained hypothesis, not a data-driven estimate of the true short-run discount process, and is the assumption the paper itself relaxes only informally in its discussion of required discount-rate volatility (Section 5.3).

5. Results

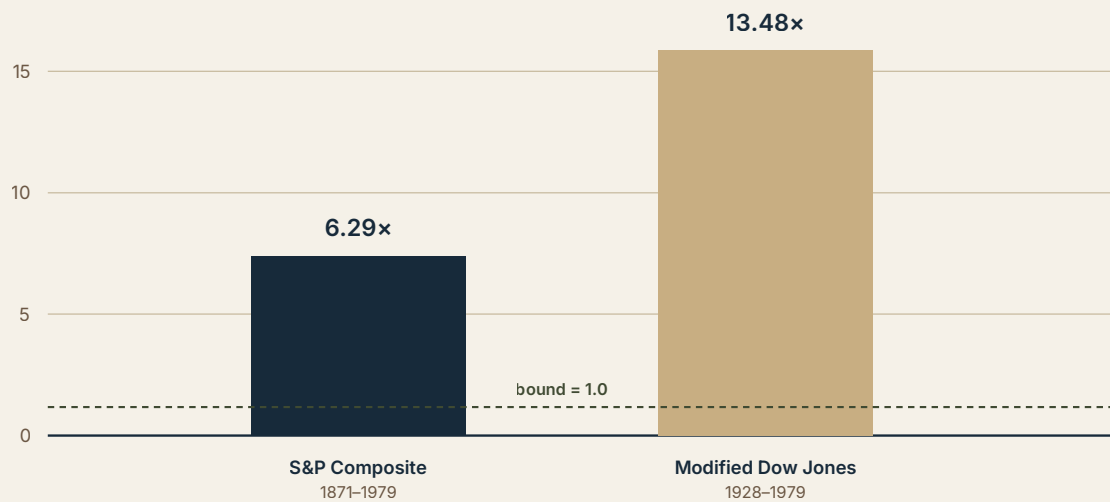
5.1 The central variance-bounds test

EVIDENCE Table 2 reports the sample standard deviations of the detrended actual price and the detrended ex-post rational price for both series, exactly as reported in the paper's Table I.

Table 2. Variance-bounds violation, detrended real series

Series	$\sigma(P_t)$	$\sigma(P_t^*)$	Ratio $\sigma(P)/\sigma(P^*)$	Discount rate r
S&P Composite, 1871–1979	47.17	7.505	6.29	4.52%
Modified Dow Jones, 1928–1979	595.6	44.18	13.48	4.29%

Source: Shiller (1981), Table I. Under the null of equation (1), the ratio in the third column should not exceed 1.

Figure 1. Actual price volatility relative to the efficient-markets bound

Bars show $\sigma(P_t)/\sigma(P_t^*)$; the horizontal line at 1.0 marks the upper bound implied by equation (1). Source: computed from Shiller (1981), Table I; own figure.

EVIDENCE Both ratios are far above the theoretical ceiling of 1.0 implied by equation (1); actual detrended prices are between roughly six and thirteen and a half times more volatile than the present value of subsequently realized dividends. The paper's own qualitative description — that the ex-post rational price series is remarkably smooth and stable relative to the actual price series — is a direct visual restatement of Table 2: a benchmark constructed from the dividend stream alone moves gently and persistently upward with the growth trend, while the actual price series exhibits large multi-year swings, including the 1920s boom and the 1930s collapse, that leave almost no trace in subsequent dividends.

5.2 Economic significance

CRITIQUE This is not a marginal statistical rejection. A ratio of 6.3 means the *variance* gap (proportional to the square of the standard-deviation ratio) is on the order of forty times too large for the S&P series, and roughly one hundred and eighty times too large for the Dow series. There is no plausible measurement-error correction, sampling-window choice, or small perturbation of the discount rate that closes a gap of this size within the model as specified; the magnitude is the paper's real contribution, because it rules out the possibility that the result is an artifact of a borderline test statistic.

5.3 Reconciling with time-varying discount rates

PAPER If r is allowed to vary over time rather than being held constant, equation (1) no longer holds in its simple form, and the paper asks how much discount-rate volatility would be needed to make the two sides of the (generalized) inequality match. The implied minimum standard deviations are large: at least 5.32 percentage points for the S&P series and 7.22 percentage points for the Dow series, implying a plausible real-rate range of roughly –6.1% to +15.2% for the S&P case and –10.1% to +18.7% for the Dow case, if real rates are assumed to move within about two standard deviations of their mean.

INTERPRETATION

A required standard deviation of real discount rates in the five-to-seven-percentage-point range, with an implied range spanning double-digit negative real rates, is far outside anything documented in short-term real interest-rate or realized-return data for the period, which is why Shiller treats a constant, or even a plausibly-varying, discount rate as unable to rescue the present-value model on its own. Section 7 revisits whether later evidence on time-varying expected returns narrows this gap.

5.4 Dividends versus earnings as the discounted variable

PAPER The paper explicitly defends discounting dividends rather than earnings: retained earnings that are reinvested show up twice under an earnings-based present-value calculation — once in the year they are earned and retained, and again, compounded, in the higher earnings they generate in future years — so an earnings-based ex-post rational price would overstate the correct present value unless a careful retention adjustment is made. Dividends are the correct terminal cash flow to the shareholder in the standard infinite-horizon valuation identity precisely because they do not double-count reinvestment.

6. Robustness and Additional Tests**CONCERN → ALTERNATIVE INDEX CONSTRUCTION**

Could the S&P result be an artifact of one index methodology? **Test:** the qualitatively identical exercise is repeated on the independently constructed, differently weighted Dow Jones series over a different, shorter sample. **Result:** the violation is not only present but larger (ratio 13.5 versus 6.3). **Resolves the concern:** partially — the direction and order of magnitude of the anomaly is robust across two independently built indices, though both share the same detrending methodology, so this does not address concerns about the detrending procedure itself.

CONCERN → DISTRIBUTIONAL SHAPE AND INFORMATION LUMPS

Could infrequent, large information arrivals (high kurtosis in dividend or price innovations) generate an apparent violation without a true efficiency failure? **Test:** the paper shows algebraically that the variance-decomposition step behind equation (1) uses only first and second moments and the orthogonality of the forecast error — it does not depend on the innovations being Gaussian. **Result:** a more leptokurtic dividend-innovation process raises $\text{Var}(u_t)$ and hence $\text{Var}(P_t^*)$, which can only widen, not narrow, the gap implied by (1). **Resolves the concern:** yes, on the paper's own terms — though this argument itself presumes the orthogonality condition holds exactly, which is the assumption later work on nonstationarity disputes.

CONCERN → SENSITIVITY TO THE DISCOUNT-RATE ESTIMATE

Is the size of the violation sensitive to the specific value used for r ? **Test:** the paper reports the required discount-rate volatility needed to reconcile the model under alternative constant- r assumptions. **Result:** because $\sigma(P_t^*)$ is not strongly sensitive to small changes in r at realistic magnitudes (the discount factor γ enters as a smoothly decaying weight), moderate perturbations of r do not close a gap of the size documented in Table 2. **Resolves the concern:** largely, for perturbations within any historically observed range of real interest rates; it does not resolve the more radical possibility of risk premia moving over a much wider range than historical short-rate data would suggest, which is exactly the possibility later exploited by the time-varying-expected-returns literature (Section 9).

CRITIQUE What the paper's own robustness section does not, and structurally cannot, test is the statistical properties of the variance-bounds estimator itself under the null hypothesis in finite, trending samples — because that critique requires simulation methods and small-sample asymptotic theory that were only developed in the years immediately following publication. This is treated as the paper's principal open robustness question in Section 7.2.

7. Critical Evaluation

7.1 What the paper actually establishes

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Conditional on a constant real discount rate, rational expectations, no bubbles, and covariance-stationary detrended series, the sample variance of detrended real U.S. equity prices exceeds the sample variance of the ex-post rational price by a large, multiply-replicated margin over 1871–1979 and 1928–1979.

SUPPORTED BUT CONDITIONAL

That this excess volatility reflects a genuine failure of the efficient-markets present-value model, rather than a statistical artifact of testing a variance inequality on trending series whose long-run growth rate must itself be estimated from the same sample.

SUGGESTIVE

That the excess volatility is evidence of investor irrationality (fads, fashions, herd behavior) rather than rational but highly time-varying risk premia; the paper raises this possibility but its variance-bounds design cannot, by itself, distinguish the two, since both would show up as a violation of the constant-discount-rate inequality.

NOT ESTABLISHED

That the mispricing identified is exploitable, that it implies predictable excess trading profits net of risk and transaction costs, or that any particular behavioral mechanism generates it; the paper does not attempt a trading-strategy or welfare calculation.

7.2 The central methodological weakness

CRITIQUE The single most important weakness, identified by a cluster of papers over 1983–1988 rather than by Shiller himself, is that the variance-bounds test is not statistically well-behaved when the underlying dividend and price series are nonstationary, or stationary only after a trend is removed that must be estimated from the same finite sample used for the test. **Problem:** Kleidon (1986) shows formally that if the true dividend process contains a unit root (or a stochastic trend), the population moments the test relies on are not the ones the sample variances consistently estimate, and Flavin (1983) shows by simulation that even under the null of exact market efficiency, the finite-sample bias in detrended variance-bounds statistics is large enough to generate apparent violations of the magnitude Shiller reports with high probability. **Why it matters:** the entire evidentiary weight of the paper rests on a single test statistic computed on two long but ultimately non-repeated historical paths; if that statistic is biased toward rejection under the null, the paper's headline finding is not wrong so much as unquantified — the true size of the excess-volatility effect, if any, is not what Table 2 reports. **Potential bias:** systematic over-rejection of market efficiency even when efficiency holds. **How it could be tested:** Monte Carlo simulation of the test statistic's distribution under the null with a data-generating process matched to the estimated trend and persistence of the actual dividend series — precisely the exercise carried out by Marsh and Merton (1986) and Mankiw, Romer and Shapiro (1985), summarized in Section 9.

7.3 Marsh–Merton: endogenous dividends

CRITIQUE A second, independent weakness raised by Marsh and Merton (1986) is that the paper's orthogonality argument implicitly assumes dividends are set independently of the price process, whereas actual corporate dividend policy (Lintner, 1956) smooths dividends toward a target payout ratio applied to a moving average of earnings, and managers plausibly condition that target partly on the firm's own valuation. If dividends are themselves a smoothed, partly price-dependent process, the orthogonality condition behind equation (3) can fail even under full rationality, and a violation of equation (1) need not indicate mispricing at all.

7.4 Strengths

CRITIQUE **Strengths recognized by the author:** the paper is explicit that its test is more demanding than, and structurally different from, standard serial-correlation tests, and it anticipates the time-varying-discount-rate rebuttal well enough to quantify how large the required variation would have to be. **Strengths identified independently:** the test requires no model of investor information sets or forecasting rules, only realized dividends, which sidesteps an entire class of joint-hypothesis problems that plague trading-rule-based efficiency tests; the finding is large enough in magnitude to survive a reasonable range of measurement error; and the use of two independently

constructed century-scale indices, rather than one, is unusually disciplined data practice for the period. The paper's clarity of exposition — reducing the entire empirical question to a single, intuitively graspable inequality between two standard deviations — is itself a methodological contribution that shaped how the profession communicates present-value tests.

7.5 Weaknesses beyond the central one

- **Constant discount rate as maintained hypothesis, not tested restriction.** The paper treats a constant r as the natural null and time variation in r as the escape hatch to be argued against qualitatively, rather than nesting both within a single estimable model.
- **Small effective sample size for a century-scale claim.** Annual data over roughly a century yields on the order of 100 independent-looking observations, but persistence in both price and dividend series sharply reduces the effective degrees of freedom available to pin down a low-frequency phenomenon like multi-decade excess volatility.
- **No formal standard errors on the variance ratio.** The paper reports point estimates of $\sigma(P)$ and $\sigma(P^*)$ without a sampling-distribution-based confidence interval for their ratio, leaving the reader to judge economic rather than statistical significance — a gap the subsequent econometric literature was built to fill.

8. Contradictions and Edge Cases

EXTENSION Under what conditions should the result disappear? If the true discount rate follows a highly persistent, economically-grounded process — for instance a consumption-based stochastic discount factor with strong habit formation, as in Campbell and Cochrane (1999) — a large share of the apparent excess volatility can be rational, since the model's constant- r assumption would then simply be misspecified rather than efficiency being violated. The result would also disappear, mechanically, under Kleidon's (1986) and Flavin's (1983) critique if the detrending bias alone is large enough to generate the observed gap even with exact efficiency.

EXTENSION When would the mechanism fail? The orthogonality argument behind equation (1) fails whenever dividends are set with reference to prices (Marsh and Merton's endogenous-dividend critique), whenever a rational bubble component violates the transversality condition assumed away in the derivation, or whenever the true discount-rate process is correlated with dividend innovations in a way that breaks the clean variance-additivity step.

EXTENSION Could another mechanism generate the same result? Yes, and this is the paper's central identification problem: both irrational fads (noise-trader sentiment shocks uncorrelated with fundamentals, formalized later by De Long, Shleifer, Summers and Waldmann, 1990) and rational, highly volatile time-varying risk premia (Cochrane, 1991, 1992) generate exactly the same signature — a price series more volatile than the constant-discount-rate present value of dividends. The variance-bounds test, by design, cannot distinguish which of the two is operating; distinguishing them requires additional structure (a model of the stochastic discount factor, or of investor sentiment) that lies outside this paper's scope.

EXTENSION Does the empirical relationship survive different samples? Mankiw, Romer and Shapiro's (1985) bias-corrected reworking of the same test on similar data still finds excess volatility, but at a materially smaller magnitude than Shiller's original ratios, indicating the qualitative direction of the finding is more robust than its exact size. West's (1988) alternative specification test, designed to be robust to some of the small-sample critiques, likewise continues to reject the simple constant-discount-rate model, which is the strongest surviving piece of evidence that some real phenomenon, not pure estimation bias, underlies the original result.

EXTENSION What happens at extreme values? In the limiting case where dividends are a pure random walk with no mean reversion, the detrending step used to obtain a stationary series is itself only an approximation, and the variance-bounds test degenerates — both $\sigma(P)$ and $\sigma(P^*)$ grow without an obviously meaningful stationary variance to compare, which is precisely the regime Kleidon's critique targets. At the opposite extreme, if dividends were perfectly smooth (zero variance, as with an idealized fixed-coupon perpetuity), the ex-post rational price would be nearly constant, and the bound in equation (1) would trivially imply that no rational asset-pricing model could tolerate the observed price volatility at all — an extreme version of exactly the tension the paper documents in the actual data.

9. Further Literature

Fama (1970) — “Efficient Capital Markets: A Review of Theory and Empirical Work”

RELATIONSHIP Direct predecessor; supplies the weak/semi-strong/strong-form efficiency taxonomy and the martingale characterization that Shiller's paper argues is too weak a standard, since a martingale price can still be arbitrarily detached from fundamental value.

LeRoy and Porter (1981) — “The Present-Value Relation: Tests Based on Implied Variance Bounds,” *Econometrica*

RELATIONSHIP Simultaneous, methodologically parallel paper using an earnings-based variance-bounds test; the two papers jointly define the excess-volatility research program and are almost always cited together.

Grossman and Shiller (1981) — “The Determinants of the Variability of Stock Market Prices,” *AER Papers and Proceedings*

RELATIONSHIP Companion piece the same year, asking whether a consumption-based marginal-utility discount factor can rationalize the observed volatility — an early bridge toward the equity-premium literature.

Flavin (1983) — “Excess Volatility in the Financial Markets: A Reassessment of the Empirical Evidence,” *Journal of Political Economy*

RELATIONSHIP Contradicts the size of the original finding; documents substantial small-sample bias in detrended variance-bounds statistics, the first serious statistical counter-critique.

Kleidon (1986) — “Variance Bounds Tests and Stock Price Valuation Models,” *Journal of Political Economy*

RELATIONSHIP Contradicts the statistical validity of the test itself under nonstationary dividends; the most cited methodological critique of this paper.

Marsh and Merton (1986) — “Dividend Behavior for the Aggregate Stock Market,” *Journal of Business*

RELATIONSHIP Contradicts the exogenous-dividend assumption underlying the orthogonality argument, using the Lintner (1956) smoothed-dividend model.

Mankiw, Romer and Shapiro (1985) — “An Unbiased Reexamination of Stock Market Volatility,” *Journal of Finance*

RELATIONSHIP Improves identification by correcting the small-sample bias raised by Flavin and Kleidon; finds a smaller but still statistically significant excess-volatility effect — a partial replication.

West (1988) — “Dividend Innovations and Stock Price Volatility,” *Econometrica*

RELATIONSHIP Improves identification with a specification test robust to several of the earlier statistical objections; continues to reject the constant-discount-rate model.

Campbell and Shiller (1987, 1988) — cointegration and the log dividend–price ratio, *Journal of Political Economy* and *Review of Financial Studies*

RELATIONSHIP Direct extension by one of the same authors; recasts the present-value relation in a log-linear, cointegration-robust form that addresses the nonstationarity objection head-on and links excess volatility to predictable returns via the dividend–price ratio.

Cochrane (1991, 1992) — time-varying expected returns and the decomposition of price–dividend-ratio variance

RELATIONSHIP Later development; reframes excess volatility primarily as evidence of time-varying discount rates (return predictability) rather than irrationality, using variance-decomposition methods that trace directly back to this paper's logic.

De Long, Shleifer, Summers and Waldmann (1990) — “Noise Trader Risk in Financial Markets,” *Journal of Political Economy*

RELATIONSHIP Applies the excess-volatility intuition in a formal behavioral model, showing sentiment-driven noise traders can generate persistent, risk-relevant price volatility unrelated to fundamentals — a rigorous behavioral alternative to the rational-time-varying-discount-rate story.

10. Research Gaps

Paper-Level Gaps

GAP

No formal sampling distribution for the variance ratio; the paper reports point estimates without a confidence interval that would allow the reader to judge statistical, as opposed to purely visual, significance.

WHY IT MATTERS

Without it, the size of the true anomaly (if any) net of estimation noise cannot be pinned down from this paper alone.

POSSIBLE WAY FORWARD

Bootstrap or Monte Carlo simulation of the test statistic under a dividend-process null calibrated to the actual sample, as later carried out by Flavin (1983) and Mankiw, Romer and Shapiro (1985).

Literature-Level Gaps

GAP

Thirty-plus years after the original debate, the literature has not fully separated how much of measured excess volatility is rational time-varying risk premia versus behavioral mispricing, because both are observationally similar in aggregate price and dividend data alone.

WHY IT MATTERS

The policy and welfare implications of the two explanations are entirely different — one implies markets are functioning but reward-for-risk is volatile, the other implies a correctable inefficiency.

POSSIBLE WAY FORWARD

Combine aggregate variance-bounds evidence with survey-based measures of investor expectations (e.g., the expectations data used in later Shiller work) that can directly separate expected-return variation from sentiment.

Methodological Gaps

GAP

Better small-sample inference for low-frequency, highly persistent macro-financial series remains an open problem in applied time-series econometrics generally, not just for this test.

WHY IT MATTERS

Any century-scale test of a slow-moving hypothesis is inherently starved of independent observations, however long the calendar span.

POSSIBLE WAY FORWARD

Panel extensions across multiple national equity markets (addressed partially by later international excess-volatility studies) to gain cross-sectional, not just time-series, identifying variation.

11. Research Opportunities

Idea 1 — Bias-corrected variance bounds across national equity markets, 1900–2025 (Extension / New Country)

RESEARCH QUESTION Does the Mankiw-Romer-Shapiro bias-corrected variance-bounds ratio show a consistent, internationally common excess-volatility magnitude across major developed equity markets, or does it vary systematically with measures of market liquidity and institutional development?

MOTIVATION The original test has been run almost exclusively on U.S. data; a genuinely cross-country panel would separate a universal feature of equity pricing from a U.S.-specific historical artifact.

HYPOTHESIS The bias-corrected ratio is significantly above 1 in every market examined but is systematically smaller in markets with higher historical trading liquidity.

ECONOMIC MECHANISM Liquidity and the depth of arbitrage capital limit how far time-varying sentiment or discount-rate shocks can push prices from the present-value benchmark.

LITERATURE CONNECTION Mankiw, Romer and Shapiro (1985); Campbell and Shiller (1988); international asset-pricing literature on liquidity premia.

DATA Global Financial Data or Jordà-Schularick-Taylor long-run cross-country equity and dividend series.

VARIABLES Detrended real price and dividend indices per country; a liquidity proxy (turnover, bid-ask spread history where available).

METHOD Bias-corrected variance-bounds estimator with bootstrapped confidence intervals, cross-sectionally regressed on liquidity proxies.

IDENTIFICATION STRATEGY Cross-country variation in liquidity development, controlling for common global shocks via time fixed effects.

EXPECTED CONTRIBUTION First systematic cross-country decomposition of excess volatility into a liquidity-related and a liquidity-independent component.

MAIN RISK Long-run dividend and liquidity data of comparable quality may not exist for enough markets to identify the cross-sectional relationship precisely.

Idea 2 — Separating risk-premium and sentiment components with survey expectations (Extension / New Methodology)

RESEARCH QUESTION How much of the variance gap documented in this paper is attributable to time-varying, survey-measured expected returns versus a residual unexplained by any rational expectations proxy?

MOTIVATION The original test cannot separate rational risk-premium variation from irrational sentiment; survey expectations data (unavailable to Shiller in 1981, but extensively collected since) offer a direct, if imperfect, window into investor beliefs.

HYPOTHESIS Survey-based expected-return variation accounts for a minority, but non-trivial, share of the variance gap, leaving a statistically significant unexplained residual.

ECONOMIC MECHANISM If investors' own stated return expectations do not move enough to rationalize observed price swings, the residual variance is difficult to attribute to a rational discount-rate story.

LITERATURE CONNECTION Greenwood and Shleifer (2014) on expectations and returns; Cochrane (1991, 1992); the original variance-decomposition logic of this paper.

DATA Survey of Professional Forecasters, UBS/Gallup or Yale investor confidence survey series, matched to aggregate dividend–price data.

VARIABLES Survey-implied expected return, realized ex-post rational price, detrended actual price.

METHOD Variance decomposition attributing $\text{Var}(P_t)$ to a survey-expectations-implied component and a residual, following the Campbell-Shiller log-linear framework.

IDENTIFICATION STRATEGY Uses the timing and revision structure of survey expectations as the source of identifying variation, rather than assuming rational expectations a priori.

EXPECTED CONTRIBUTION A direct, non-model-dependent estimate of how much of historical excess volatility is sentiment-driven.

MAIN RISK Survey expectations are a noisy and possibly non-representative proxy for the marginal investor's true beliefs, limiting the strength of any causal claim.

Idea 3 — A variance-bounds test for private, illiquid asset classes (New Asset Class)

RESEARCH QUESTION Do appraisal-based valuations in private real estate or private equity exhibit the same or a materially different excess-volatility signature relative to realized cash-flow present values as public equities?

MOTIVATION Private-market valuations are smoothed by appraisal practices, so the same test could reveal an opposite bias (too little apparent volatility) and shed light on how much of listed-equity excess volatility is a liquidity/trading phenomenon.

HYPOTHESIS Appraisal-based series show a variance ratio below, not above, 1, consistent with smoothing rather than excess volatility.

ECONOMIC MECHANISM Infrequent trading and appraisal-based marking artificially dampens measured price variance relative to the true, unobserved transaction-price variance.

LITERATURE CONNECTION Geltner (1991) on appraisal smoothing in real estate indices; this paper's variance-bounds logic applied in reverse.

DATA NCREIF property-level appraisal data and cash-flow (net operating income) histories; private-equity cash-flow data from Preqin or Burgiss.

VARIABLES Appraised value, realized net operating income or exit cash flows, constructed ex-post rational value.

METHOD The same variance-bounds construction as this paper, adapted for irregular cash-flow timing.

IDENTIFICATION STRATEGY Comparison of appraisal-based variance to variance implied by realized transaction prices at sale, where available, as an unsmoothed benchmark.

EXPECTED CONTRIBUTION Quantifies appraisal smoothing bias directly in variance-bounds terms, complementing the existing de-smoothing literature.

MAIN RISK Cash-flow and valuation data of sufficient length and quality for private assets are much scarcer than for century-scale public equity indices.

Idea 4 — Machine-learning-based nonparametric present-value benchmarks (New Methodology)

RESEARCH QUESTION Does a flexible, nonparametric (rather than constant-discount-rate) estimate of the ex-post rational price, learned from macro and dividend covariates, still find a significant residual excess-volatility gap?

MOTIVATION A nonparametric benchmark can absorb far more of the plausible time-variation in discount rates than the paper's original constant- r model, providing a much harder test of whether any gap remains.

HYPOTHESIS A significant, if smaller, excess-volatility gap survives even after a flexible model absorbs discount-rate variation explainable by observable macro covariates.

ECONOMIC MECHANISM If the residual gap remains large, the unexplained component is a stronger candidate for sentiment-driven mispricing than in the original constant- r test.

LITERATURE CONNECTION Gu, Kelly and Xiu (2020) on machine learning in empirical asset pricing; Cochrane (1992) variance decomposition.

DATA Shiller's own long-run S&P dataset (publicly maintained and updated), augmented with macro covariates (inflation, industrial production, term spread).

VARIABLES Detrended real price and dividends; macro covariates as discount-rate predictors.

METHOD Regularized nonparametric regression (e.g. gradient-boosted trees or a shrinkage estimator) to model time-varying r_t , embedded in the present-value recursion.

IDENTIFICATION STRATEGY Out-of-sample cross-validation to avoid overfitting the discount-rate model to the very price data the test is meant to evaluate.

EXPECTED CONTRIBUTION A modern, flexible re-test of the paper's core claim that goes well beyond the constant-discount-rate null without abandoning the variance-bounds logic.

MAIN RISK A sufficiently flexible model risks absorbing genuine mispricing into the fitted discount-rate process, mechanically shrinking the residual gap without a real economic justification.

Idea 5 — Replication with post-1979 out-of-sample data and structural-break testing (Replication)

RESEARCH QUESTION Does the original variance-bounds violation persist, in bias-corrected form, over the 1980–2025 period, and is there a statistically detectable structural break in the ratio around major volatility episodes (1987 crash, 2000 dot-com peak, 2008 crisis)?

MOTIVATION A genuine out-of-sample extension using data unavailable to the original author is the cleanest possible test of whether the phenomenon is a persistent feature of equity markets.

HYPOTHESIS The bias-corrected ratio remains significantly above 1 in every subperiod, with local maxima around the identified bubble/crash episodes.

ECONOMIC MECHANISM Episodic sentiment or leverage-driven price swings around known bubble episodes should show up as local spikes in the variance ratio.

LITERATURE CONNECTION Mankiw, Romer and Shapiro (1985) bias correction; Shiller's own updated long-run dataset and *Irrational Exuberance* (2000) narrative episodes.

DATA Shiller's publicly maintained long-run S&P 500 price, dividend and earnings dataset through the present.

VARIABLES Detrended real price and dividends, sub-sample and rolling-window variance ratios.

METHOD Bias-corrected variance-bounds estimator applied on rolling and structural-break-tested subsamples (Bai-Perron break tests).

IDENTIFICATION STRATEGY Structural-break dating against independently documented bubble/crash episodes, avoiding data-mined break dates.

EXPECTED CONTRIBUTION A direct, modern replication using forty-five additional years of data and the bias corrections the original paper lacked.

MAIN RISK Structural breaks in dividend policy (the post-1980s shift toward share buybacks over dividends) complicate direct comparability with the original 1871–1979 dividend-based construction.

Ranking

CRITIQUE Ranked by novelty, feasibility, data availability, methodological quality, economic importance and expected contribution, the top three are, in order: **Idea 5** (out-of-sample replication with bias correction and structural breaks), because Shiller's own dataset is already publicly maintained through the present, data availability is excellent, and a clean out-of-sample test speaks most directly to whether the original 1981 finding was a historical artifact; **Idea 2** (survey-expectations decomposition), because it attacks the paper's central unresolved identification problem — rational risk premia versus sentiment — more directly than any purely price-and-dividend-based re-test can; and **Idea 1** (cross-country bias-corrected replication), because it is the most feasible large-sample extension and directly tests whether the phenomenon is universal or U.S.-specific, a question the original single-country design cannot answer. Ideas 3 and 4 are promising but constrained respectively by data scarcity and by the risk of the flexible model absorbing the very effect it is meant to detect.

12. Learning Framework

Prerequisite Concepts

- The dividend-discount / Gordon growth model and the general infinite-horizon present-value identity for asset prices.
- Conditional expectation as a variance-minimizing (orthogonal) projection, and the basic variance-decomposition identity $\text{Var}(X) = \text{Var}(E[X|I]) + \text{Var}(X - E[X|I])$.
- Covariance-stationarity and why sample variances are only meaningful estimators of population variances under it.
- The martingale/random-walk characterization of efficient prices (Fama, 1970; Samuelson, 1965), to see how this paper's test differs from it.

Core Concepts Learned

- How to construct a variance-bounds test as an alternative to return-predictability tests of market efficiency.
- The distinction between testing whether price *changes* are predictable and testing whether the *level* of price is consistent with a present-value model.
- Why detrending is necessary, and dangerous, when the trend itself must be estimated from the test sample.
- The observational-equivalence problem between rational time-varying discount rates and behavioral mispricing.

Important Equations

- Equation (2): the constant-discount-rate present-value pricing identity.
- Equation (4): the ex-post rational price, built from realized rather than expected dividends.
- Equation (1): the variance-bounds inequality $\sigma(P_t) \leq \sigma(P_t^*)$, the paper's entire testable content.

Methods Learned

- Variance-decomposition testing of rational-expectations models using realized outcomes as an ex-post benchmark.
- Detrending long macro-financial time series to approximate covariance-stationarity before variance comparison.
- Back-of-envelope calibration of how much an auxiliary parameter (here, discount-rate volatility) would need to move to rescue a rejected model.

Questions to Test Understanding

COMPREHENSION

Can I state, without notes, why a rational conditional expectation must have weakly lower variance than the random variable it forecasts, and why that fact alone is the entire content of equation (1)?

ANALYSIS

Can I explain why detrending using a trend estimated from the same sample can bias the variance-bounds test toward rejecting efficiency even when efficiency holds, and reconstruct informally why Kleidon's nonstationarity critique follows from this?

RESEARCH

Can I design a test that would distinguish rational time-varying discount rates from sentiment-driven mispricing as the source of a measured variance-bounds violation, using data not available to Shiller in 1981?

Papers to Read Next

- LeRoy and Porter (1981), “The Present-Value Relation: Tests Based on Implied Variance Bounds,” *Econometrica*.
- Kleidon (1986), “Variance Bounds Tests and Stock Price Valuation Models,” *Journal of Political Economy*.
- Campbell and Shiller (1988), “The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors,” *Review of Financial Studies*.

13. Conclusion

One-Sentence Thesis

Detrended U.S. real stock prices, 1871–1979, are between six and thirteen times more volatile than the present value of subsequently realized dividends permits under a constant-discount-rate, rational-expectations model, a gap too large to be explained away by plausible discount-rate movement or fat-tailed information arrival, though its ultimate cause — statistical bias, rational time-varying risk premia, or investor sentiment — remained unresolved by this paper alone.

Three Most Important Insights

- A market can satisfy every standard test of return unpredictability and still be persistently detached, in level, from the present value of its own future cash flows.

- A variance-bounds test built from realized outcomes avoids modeling investors' information sets directly, but inherits new fragilities of its own — principally sensitivity to nonstationarity and to the assumption that dividends are set independently of prices.
- Excess volatility is observationally consistent with two very different worlds — rational but highly variable discount rates, or behavioral mispricing — and separating them requires evidence well beyond aggregate price and dividend data.

Strongest Evidence

The consistency of the direction and rough order of magnitude of the violation across two independently constructed indices (S&P Composite and modified Dow Jones) spanning different, only partially overlapping historical periods.

Biggest Weakness

The absence of a formal small-sample sampling distribution for the variance ratio, which left the test vulnerable to the Flavin (1983) and Kleidon (1986) demonstrations that detrending bias alone can produce apparent violations of a similar order of magnitude under the null of efficiency.

Most Important Research Gap

No design in this literature cleanly separates rational time-varying risk premia from sentiment-driven mispricing as the source of the residual, bias-corrected excess volatility that later work (Mankiw, Romer and Shapiro, 1985; West, 1988) continued to find.

Best Research Opportunity

An out-of-sample, bias-corrected replication using Shiller's own publicly maintained long-run dataset through the present, with structural-break testing around documented bubble and crash episodes (Idea 5, Section 11).

Three Papers to Read Next

- LeRoy and Porter (1981), *Econometrica*.
- Kleidon (1986), *Journal of Political Economy*.
- Campbell and Shiller (1988), *Review of Financial Studies*.

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